1. Inventory Management

1.1. Inventory

- Materials needed for the production of goods and services.

Types:

1. Raw materials
   - Purchased from outside suppliers
2. Work in progress
   - Work in progress is any product which is not yet converted into finished goods.
   - Depends on time period of production and production method used.
3. Finished goods
   - Good ready to be sold to consumers

1.2. Inventory management

- Without effective inventory management, there maybe many problems.
- Insufficient inventories to meet unforeseen changes in demand
- Out-of-date inventories maybe held
- Wastage due to incorrect storage conditions
- High storage and opportunity costs if extra inventory is held
- Poor management may lead to delayed deliveries, ignoring discounts, etc

1.3. Inventory holding

Inventory holding costs

1. Opportunity cost – working capital tied up in inventory could be used elsewhere. Higher interest rates, higher the opportunity cost of holding inventory.
2. Storage cost – inventories must be held in appropriate, safe conditions to avoid wastage. Higher inventory, higher the storage costs
3. Risk of wastage and obsolescence – if inventories are kept unused, they may become obsolete, lowering the value of such inventories and increasing the business's expenses

Costs of not holding inventories

1. Lost sales – business may not be able to supply all customers, leading to loss of sales and revenue
2. Idle production resources – if raw material inventories run out, expensive equipment and workers will be left idle, leading to loss of output and wasted resources.
3. Special orders could be expensive – urgent orders given to suppliers may lead to extra delivery, administration costs
4. Small order quantities – higher average costs as the company will not benefit from economies of scale

1.4. Optimum order size

- Operations managers must ensure they have enough stocks to allow for smooth production.
- Operations managers usually order their inventories on the basis of the Economic Order Quantity (EOQ)
- EOQ is the optimum inventory level where the costs incurred are minimum (both re-ordering costs and stock-holding costs)

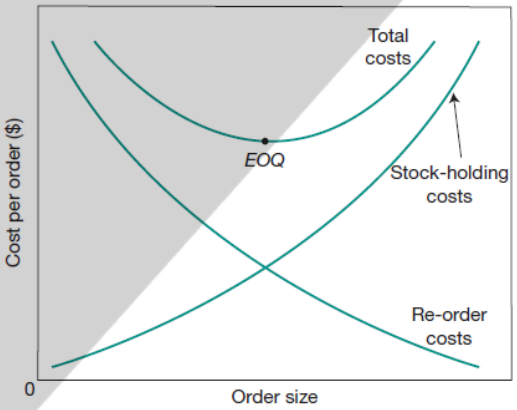

Figure 24.2 Factors influencing the economic order quantity

- The reorder costs decrease as the order size increases, showing a downward sloping graph.
- Whereas, the stock-holding costs increase as the order size rises, showing an upward sloping graph.
- The Economic Order Quantity is shown where the stock-holding costs and re-order costs curves intersect.

1.5. Inventory control graphs

- Inventory-control graphs record the buffer inventories, maximum inventory.
- They help in determining the right order time and quantity.
- Buffer inventory – minimum inventory held to deal with delays in delivery and unforeseen demand changes
- Maximum inventory level – the maximum quantity of inventory the company can hold, space and financial terms
- Reorder quantity – the number of units ordered each time
- Lead time – time taken for the supplier to deliver the raw materials
- Reorder stock level – the level of inventory which will trigger a new order

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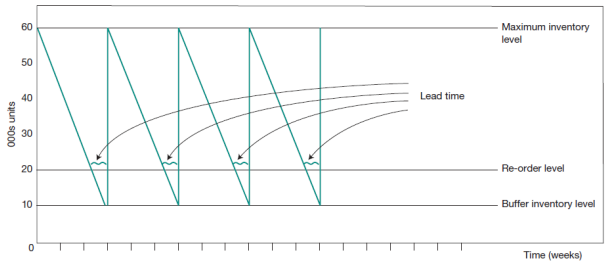

Figure 24.3 A typical inventory-control chart

The maximum inventory is 60000 units. The buffer inventory level is 10000 units and the reorder level is 20000 units. The reorder quantity is 50000 units (60000-10000)

1.6. Just-in-Case (JIC) Inventory Control

This management system targets to lower the risk of running out of inventory to the minimum by holding high buffer inventory levels.

Benefits of JIC inventory management	Drawbacks of JIC inventory management
JIC makes it highly unlikely to run out of inventory, so production can continue even with major supply delays.	High capital cost as finance is required to invest in inventories
Lower requirement to forecast sales as accurately as with JIT due to high inventory level	Other costs such as space, holding and other variables associated with inventory, increasing overall expense of business
Takes advantage of purchasing economics of scale by ordering supplies in large quantities, lowering average supply cost	Inventories are highly likely to lose its value due to potential change in fashion or technology

1.7. Supply Chain Management

- Supply chain: the network of all the business activities involved in creating a product for sale, starting with the delivery of raw materials and finishing with the delivery of the finished product.

  supply chain management: Management of the entire production flow of a product (from raw materials to finished product) to minimise costs as well as improve customer service.

Supply chain management targets to decrease the time period of converting inputs into outputs.

Supply management system shorten the time period by:

- Build sturdy and consistent communication with suppliers to ensure on-time and high-quality delivery of raw materials.
- Improve transport systems to speed up delivery.
- Speed up new product development to stay competitive in the market
- Use technology and flexible workforces to speed up production.
- Reduce waste at all stages to cut costs.

Benefit of effective supply chain management:

- It can create happier customers by providing high quality customer service by ensuring fast, on-time delivery of quality products.
- Reducing expenses on buying, storing, and producing goods by saving time and decreasing waste will lower operating cost.
- Improvement in inventory management and efficiency, will inevitably lead to higher business profits.

1.8. Just in time (JIT) inventory control

- It is an inventory control system which avoids the need to hold inventories. They arrive just as and when required

Requirements for JIT

- Excellent relations with supplier
- Flexible and multiskilled production staff
- Flexible equipment and machinery
- Accurate demand forecasts
- Latest IT technology
- Excellent employee-employer relations
- Quality must be priority

Advantages and disadvantages

Advantages	Disadvantages
■ Capital invested in inventory is reduced and the opportunity cost of inventory holding is reduced. ■ Costs of storage and inventory holding are reduced. Space released from holding of inventories can be used for a more productive purpose. ■ Much less chance of inventories becoming outdated or obsolescent. Fewer goods held in storage also reduces the risk of damage or wastage. ■ The greater flexibility that the system demands leads to quicker response times to changes in consumer demand or tastes. ■ The multiskilled and adaptable staff required for JIT to work may gain from improved motivation.	■ Any failure to receive supplies of materials or components in time caused by, for example, a strike at the supplier's factory, transport problems or IT failure will lead to expensive production delays. ■ Delivery costs will increase as frequent small deliveries are an essential feature of JIT. ■ Order-administration costs may rise because so many small orders need to be processed. ■ There could a reduction in the bulk discounts offered by suppliers because each order is likely to be very small. ■ The reputation of the business depends significantly on outside factors such as the reliability of supplying firms.

JIT evaluation

- JIT requires employees to be accountable for their performance and suppliers to be reliable
- JIT may be unsuitable when:
- Costs of halting production exceed inventory holding costs
- Expensive IT systems cannot justify potential cost savings
- Global inflation makes holding inventories cheaper

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